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Hedging at Porsche

When Porsche made headlines, it was usually for the engineering of the latest edition of one of its famous sports cars—not for complex financial engineering. But the results presented at the company's annual press conference on November 28, 2007, stunned the business world. While Porsche made a respectable profit of around €1bn[i] from manufacturing and selling cars, this number was dwarfed by the huge profits Porsche made from transactions in financial derivatives. Porsche had undertaken large trades in the foreign exchange options market that paid off handsomely as the U.S. dollar weakened. Moreover, Porsche had bought a huge number of options on shares of Volkswagen, Europe's biggest automaker. These trades together added roughly €4 billion of profit to Porsche's bottom line.[1]

Analysts were divided in their reaction when Holger Härter, Porsche's CFO, presented these stunning results. Some went as far as to describe Härter as a "financial genius." Others questioned the wisdom of an industrial company engaging in derivatives trades of this scale. Some analysts remarked wryly that Porsche had become a "hedge fund."[2]

Did Porsche simply earn the fruits of prudent and clever risk management? Or did these results indicate that Porsche's executives engaged in reckless financial policies that exposed the company to outsized risks?

Background

With headquarters in Stuttgart, Germany, Porsche was a manufacturer of performance sports cars. Its product line in 2007 included the 911, Boxster, and Cayman sports cars, and the Cayenne sports utility vehicle. The four-door Panamera was planned to be launched in 2009. The production of the 911 and Cayenne vehicles took place at the company's German plants in Stuttgart and Leipzig, while much of the assembly of the Boxster and Cayman was outsourced to the Finnish company Valmet.[3]

The company's sales were heavily dependent on the U.S. and German markets. Each one of these two markets accounted for about a third of total sales in the fiscal year 2006/07.[4]

i The exchange rate at the end of November 2007 was 1.47 $ per €.

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In the early 1990s Porsche had been on the brink of bankruptcy. At the time, sales were heavily dependent on a single product: the 911 sports car. As recession hit the United States and Germany, sales of expensive sports cars slumped. Combined with the effects of a weak U.S. dollar, this had a devastating effect on Porsche. Sales fell dramatically from more than 50,000 a year to 14,000 units in 1993.[5]

In 1993, then 40-year-old Wendelin Wiedeking was appointed as CEO to lead Porsche out of this crisis. Under his leadership the company brought in former Toyota engineers from Japan to adopt lean manufacturing techniques, and it focused on the development of the Boxster, a new entry-level model priced considerably below the 911.[6]

In a relatively short time, Porsche completed a successful turnaround. Wiedeking was still CEO in 2007. By that time Porsche had achieved the highest profit margins in the industry, earning itself the label "most profitable car company in the world."[7]

Porsche was essentially a privately held company. The company had two classes of shares: ordinary shares with voting rights and preferred nonvoting shares. All of Porsche's 8.75 million ordinary shares were held by the Piëch and Porsche families. The 8.75 million (nonvoting) preferred shares were publicly traded and the majority held by institutional investors.[8]

Risk Management Policy

Porsche's risk management policy was strongly influenced by its experience of nearly going bankrupt in the early 1990s. Its policy was grounded in several lessons Porsche's executives learned from this period of trouble.

First, Porsche aimed to have little leverage on its balance sheet and a cushion of ample liquidity. The firm had little long-term debt. Its financial liabilities, which consisted mostly of bank loans and bonds, accounted for only a small fraction of total assets. It also maintained a balance of cash and highly liquid cash-equivalent assets of more than €2bn (see **Appendix A**). Instead of relying on credit lines provided by banks, the company preferred to keep significant cash balances. This cautious policy also had its roots in Porsche's troubles in the early 1990s. When the crisis hit back then, the company struggled as banks became unwilling to extend credit. As Porsche CFO Holger Härter put it, "We learned the hard way that banks are there for you when you don't need them, and when you do need them, they're nowhere to be seen."[9]

Second, Porsche aimed to hedge, to a large extent, its foreign exchange exposure. During the troubles of the early 1990s Porsche was not well hedged against a falling U.S. dollar, which compounded the problems the company faced at the time. Its German competitors BMW and Mercedes responded to this challenge by establishing and expanding production facilities in the United States. By increasing their dollar-cost base, they were trying to match dollar revenues with dollar costs. Porsche chose not to go down this route of establishing such a "natural hedge." Instead, it relied on foreign exchange derivatives to hedge its exposure to foreign exchange risk.[10]

Foreign Exchange Hedging

Porsche's foreign exchange hedging program made use of foreign exchange options. Options were used to lock in a desired floor to the exchange rate. In particular, Porsche often used at-the-money options[ii] to lock in an exchange rate close to the current spot rate.[11]

ii An option is called an at-the-money option if the price of the underlying asset equals the strike price.

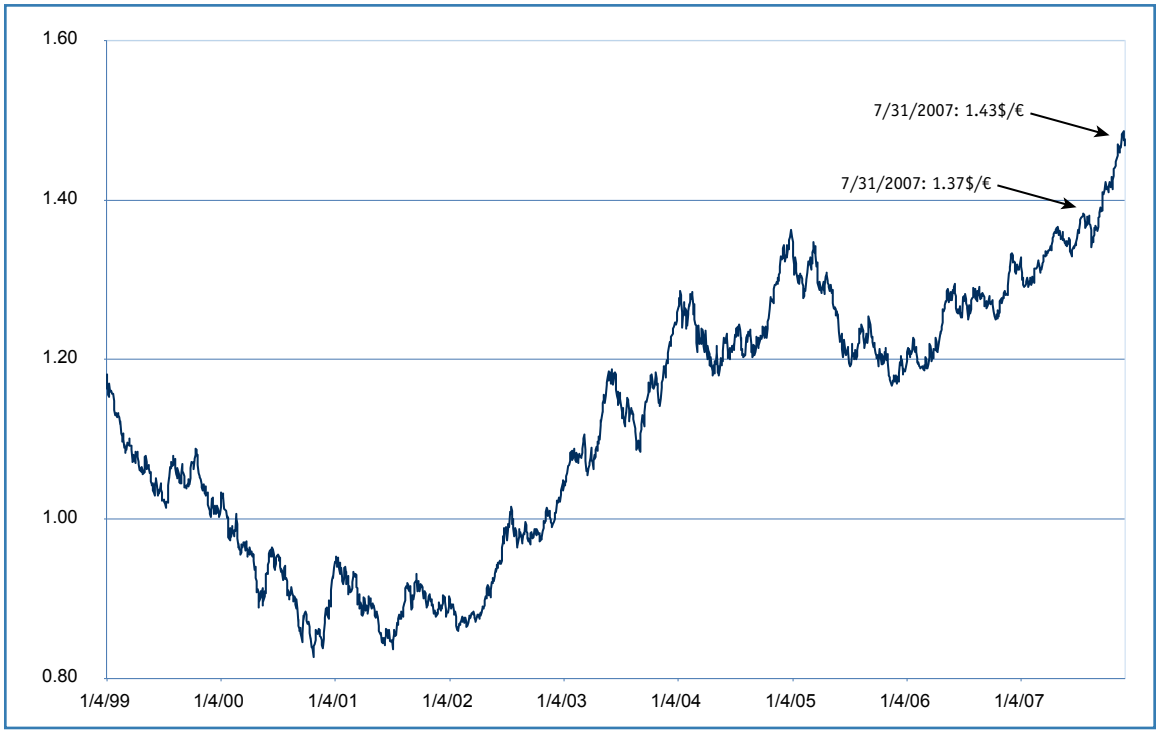
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For example, on November 30, 2007, the US$-€ exchange rate stood at a spot rate of 1.47 $ per € (see **Exhibit 1**). If Porsche bought one-year put options on the U.S. dollar, at a strike of 1.45 $ per €, it obtained the right, but not the obligation, to sell U.S. dollars one year from now at a predetermined exchange rate of 1.45 $ per €. In this way, the company locked in a minimum amount of euros it would obtain per U.S. dollar when it converted U.S. dollar revenue from sales in the United States into euros.

Exhibit 1

Dollar/Euro Exchange Rate ($ Per €)



Note: The average $/€ rate during Porsche's fiscal year 2006/07 (ending in July 31, 2007) was 1.31 $/€.

Source: DataStream.

Porsche designed its hedging program so that its forecast sales three years out would be fully hedged (against adverse movements in the exchange rate). This involved buying and rolling over a portfolio of put options with various maturities of up to three years. On July 31, 2007, Porsche held options with a notional amount of €12 billion and a market value of about €0.5 billion (see **Exhibit 2**).[12]

Assuming a simple put-option hedge and that the entire position is a U.S. dollar hedge, a notional amount of €12 bn would mean that if Porsche had exercised these put options on July 31, 2007, it would have been able to sell about €12 billion worth of dollars at a pre-determined exchange rate given by the strike of the put options. €12 billion is roughly three times Porsche's annual sales outside of Europe (see **Exhibit 3**, assuming some growth over the following three years) consistent with Porsche's stated goal of hedging exposure three years out.

Porsche believed that it should hedge its foreign exchange exposure largely irrespective of the views about future exchange rate developments that management might have. As Henrik Hänche, Porsche's treasurer, explained: "We want to avoid the behavioral finance trap, where we act in a certain way because

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we think the current rate of $1.19 to the euro will be higher or lower one month from now. We are convinced that it's not possible to beat the market."[13]

Exhibit 2

Notional amounts and market value of derivative financial instruments held by Porsche (€ thousands)

		July 31, 2007		July 31, 2006	
		Nominal volume T€	Total market value T€	Nominal volume T€	Total market value T€
Assets	Currency hedge	12,198,361	466,268	8,276,098	389,003
	Interest hedge	1,091,319	34,998	1,292,744	34,692
	Stock options	10,553,364	5,055,224	2,628,055	870,437
		23,843,044	**5,556,490**	**12,196,897**	**1,294,132**
Equity and liabilities	Currency hedge	1,357,879	13,866	1,441,224	9,572
	Interest hedge	902,830	27,118	886,812	34,818
	Stock options	13,473,485	2,445,118	2,441,025	980,716
		15,734,194	**2,486,102**	**4,769,061**	**1,025,106**

Source: Porsche, Annual Report 2006/2007. <http://www.porsche-se.com/pho/en/investorrelations/mandatorypublications/annualreport-14/download/>.

Exhibit 3

Sales and profit by region (€ millions)

	Germany	North America	Europe without Germany	Rest of the world
Sales	2,554	2,218	1,750	847
Profit before financial income and income tax	4,427	142	120	40

Source: Porsche, Annual Report 2006/2007. <http://www.porsche-se.com/pho/en/investorrelations/mandatorypublications/annualreport-14/download/>.

This approach was different from that taken by some of Porsche's competitors. For example, after having reaped considerable gains from U.S. dollar hedges as the euro rose against the dollar in the years before 2004, BMW decided to largely abandon its hedge. In January 2004, *The New York Times* reported about an interview with the head of BMW's risk management, according to which BMW believed that at the current exchange rate of $1.25 the euro was overvalued. The company's computer models suggested that the currency's correct long-term value was around $1.10. Based on this estimate, BMW opted not to hedge heavily.[14]

The Volkswagen Stake

On September 26, 2005, Porsche stunned financial markets when it announced plans to acquire a 20% stake in Volkswagen (VW), Europe's biggest carmaker. While Porsche was a small, highly profitable niche producer, VW was a mass producer struggling to improve its profitability. In addition to the VW brand, VW

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also owned the Audi and Skoda brands, among others.[15] The two companies could hardly be more different. Volkswagen sold more than 5 million vehicles per year, with sales approaching €100 billion, while Porsche sold about 90,000 vehicles per year, yielding sales of about €7 billion.[16]

Porsche's managers, however, believed that they had a good rationale for this unexpected move. They explained that their main objective was to prevent hedge funds from taking over and breaking up VW. Porsche argued that its bid for a VW stake was of critical importance because of Porsche's dependence on VW as its main supplier and as a partner in several joint projects. VW provided about 30% of the components used in Porsche cars. Porsche's Cayenne, VW's Tuareg, and VW subsidiary Audi's Q7 sports utility vehicles shared a common platform. Porsche was also cooperating with VW in the development of the platform for the Panamera, a four-door sports car to be introduced in 2009. The two companies were additionally cooperating on the development of hybrid and fuel-cell models. As a small niche manufacturer, Porsche had difficulty achieving sufficient economies of scale. It said that it needed VW as a reliable partner. A takeover or breakup of VW by some third party would threaten this relationship.[17]

Critics maintained that this looked like a return to the old days of German "corporatism," where companies and banks had cross-holdings in each other, protecting one another from hostile takeovers and shielding themselves from capital market pressures for improving profitability. Moreover, some critics saw conflicts of interest in the fact that Ferdinand Piëch, who served as CEO of VW from 1993 to 2002 and as the head of VW's supervisory board, was also a member of Porsche's supervisory board and owner of a large fraction of Porsche's voting shares.[18]

VW was a potential target for takeover and breakup, as it had been struggling for many years to achieve and maintain a satisfactory level of profitability. Many analysts argued that the company suffered from large inefficiencies. In particular, analysts criticized the apparent cross-subsidization from its successful and profitable Audi subsidiary to less profitable parts of the company and concluded that VW's breakup value was considerably higher than its value as a combined company.[19]

For a long time, however, VW had been shielded by an arcane German law (the Volkswagen Act of 1960) that limited the voting rights of any shareholder in VW to a maximum of 20% and gave the federal and regional government seats on the supervisory board of the company. In 2005, at the time when Porsche first announced its intention to buy a large stake in VW, the European Court was expected to mandate that Germany repeal this law in the near future. The European Commission argued that the law conflicted with European Union rules on free movement of capital within the single European market. And in October 2007 the European Court did indeed rule that the Volkswagen law illegally shielded VW from takeovers and it required the German federal government to amend or repeal this law.[20]

Porsche had hoped for this outcome. In October 2006, it announced its intention to acquire a further 3.9% of VW, raising its stake from then 21.2% to 25.1%. It also announced that it had already bought a sufficient number of call options on VW ordinary shares, so that it faced no price risk in this planned increase in its stake in VW.[21] On March 26, 2007, Porsche announced that it had exercised options for the acquisition of a further 3.6% of the ordinary shares of VW, which increased its stake to 30.9%. [22]

This increase of its stake above 30% triggered, by law, a requirement to make a mandatory offer for the remaining VW shares. Porsche, however, offered only the minimum price required by the law, € 100.92 per ordinary share, which was about 14% below the prevailing market price at the time, so hardly any shareholder took Porsche's offer. At the time Porsche held 27.3% of VW's ordinary shares and had options to buy another 3.7%.[23] The offer lapsed in May, and Porsche was now free to increase its stake to 50 percent, if it wished to do so.[24]

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Porsche had made extensive use of call options on VW ordinary shares to build its stake in VW. Porsche executives argued that this helped them to hedge against the risk that their attempt to buy VW shares would cause VW's stock price to rise. The exact nature of Porsche's option positions remained undisclosed. Some analysts believed that Porsche was holding options to another 60 million of VW shares in November 2007[25] (compared with 290.3 million ordinary VW shares outstanding).[26]

Is Porsche a Hedge Fund?

Porsche's foreign exchange hedges and the acquisition of the VW stake had a dramatic impact on Porsche's bottom line in the fiscal year 2006/07. On sales of about €7.4 billion, Porsche reported pretax profits of €5.9 billion, up from €2.1 billion in the previous year (see **Appendix B**). Profits from its VW stock option trades accounted for €3.6 billion of these profits. A further €1.2 billion came from Porsche's share in VW's profits, and thus "only" about €1.1 billion from its core business, although that still represented an impressive profit margin by car industry standards. In other words, Porsche made three times as much profit from trades in financial derivatives than from its business of manufacturing and selling cars.[27]

Moreover, the profit from its core business was also boosted by foreign exchange hedges that Porsche had in place. These hedges contributed €250 million to Porsche's bottom line in 2006/07, as the dollar weakened against the euro.[28]

Porsche's astonishing results drew praise, but also criticism from analysts and commentators. Some journalists likened Porsche's CFO, Härter, to a "financial genius," or characterized him as "more of an options trader than a CFO," while others warned that Porsche had just been lucky, and that things might have turned out differently at another time. Some even suggested that Porsche was more like a hedge fund than a car company. A company spokesman replied by pointing out that "hedge funds don't make cars the last time I checked."[29]

Conclusion

Porsche had earned a reputation for a cautious risk management. However, the company's enormous profits from transactions in financial derivatives led many to question whether this was still an accurate description. Should a company such as Porsche engage in derivatives transactions on this scale? Was its approach to building the Volkswagen stake consistent with sound risk management principles or an example of reckless speculation with shareholders' capital? (For additional insight, see **Appendices C-G**.)

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Appendices

Appendix A

Consolidated Balance Sheet of Porsche Group (€ thousands)

		Notes	July 31, 2007 T€	July 31, 2006 T€	
Assets	Intangible assets	(15)	263,526	250,295	
	Property, plant and equipment	(16)	1,378,435	1,178,352	
	Investments in associates	(17)	7,059,333	3,263,733	
	Other financial assets	(17)	67,584	27,755	
	Leased assets	(18)	990,979	960,650	
	Trade receivables	(20)	20,772	1,990	
	Receivables from financial services	(21)	1,321,635	1,248,750	
	Other receivables and assets	(22)	285,662	172,659	
	Receivables of taxes on income	(23)	63,598	0	
	Securities	(24)	1,014,573	713,072	
	Deferred tax assets	(10)	75,114	152,930	*
	Non-current assets		**12,541,211**	**7,970,186**	*
	Inventories	(19)	625,209	594,080	
	Trade receivables	(20)	245,136	202,981	
	Receivables from financial services	(21)	459,879	434,889	
	Other receivables and assets	(22)	5,604,442	1,399,988	*
	Receivables of taxes on income	(23)	27,262	1,306	*
	Securities	(24)	1,419,185	2,048,521	
	Cash and cash equivalents	(25)	2,410,066	1,988,550	
	Current assets		**10,791,179**	**6,670,315**	
			23,332,390	**14,640,501**	*
Equity and liabilities	Subscribed capital	(26)	45,500	45,500	
	Capital reserves	(26)	121,969	121,969	
	Revenue reserves	(26)	8,507,292	4,362,342	*
	Translation differences	(26)	– 3,712	– 1,821	
	Capital allocable to shareholders		8,671,049	4,527,990	*
	Hybrid capital	(26)	809,977	809,977	
	Minority interests	(26)	0	0	*
	Equity		**9,481,026**	**5,337,967**	*
	Pension provisions	(27)	719,476	658,743	
	Other provisions	(28)	624,234	628,512	
	Deferred tax liabilities	(10)	612,826	181,764	
	Financial liabilities	(29)	3,539,237	3,529,650	
	Trade payables	(30)	7,480	3,875	
	Other liabilities	(31)	67,007	51,219	
	Non-current provisions and liabilities		**5,570,260**	**5,053,763**	
	Tax provisions	(28)	896,643	238,026	
	Other provisions	(28)	1,161,098	1,012,522	
	Financial liabilities	(29)	3,010,024	1,280,342	*
	Trade payables	(30)	505,183	478,942	
	Other liabilities	(31)	2,708,156	1,238,939	
	Current provisions and liabilities		**8,281,104**	**4,248,771**	*
			23,332,390	**14,640,501**	*

* adjusted

Source: Porsche, Annual Report 2006/2007.

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Appendix B

Consolidated Income Statement of Porsche Group (€ thousands)

	Notes	2006/07 T€	2005/06 T€	
Continuing operations				
Sales	(1)	**7,367,876**	7,122,667	
Changes in inventories and other own work capitalized	(2)	162,217	172,967	
Total operating performance		**7,530,093**	7,295,634	
Other operating income	(3)	7,264,416	1,045,127	
Cost of materials	(4)	– 3,659,520	– 3,273,507	
Personnel expenses	(5)	– 1,264,325	– 1,037,475	
Amortization and depreciation	(15), (16), (18)	– 531,712	– 488,758	
Other operating expenses	(6)	– 4,600,099	– 1,709,318	
Profit before financial income		**4,738,853**	1,831,703	
Share of profit of associates	(7)	1,223,164	203,357	
Financial expenses	(8)	– 272,232	– 198,916	*
Financial income	(9)	167,215	192,053	*
Financial result		**1,118,147**	196,494	
Profit from ordinary activities of continuing operations		5,857,000	2,028,197	
Profit from ordinary activities of discontinued operations		0	81,803	
Profit from ordinary activities		**5,857,000**	2,110,000	
Income taxes from continuing operations	(10)	– 1,615,000	– 713,578	
Income taxes from discontinued operations	(10)	0	– 3,422	
Income taxes	(10)	– 1,615,000	– 717,000	
Net profit from continuing operations		4,242,000	1,314,619	
Net profit from discontinued operations	(11)	0	78,381	
Net profit		**4,242,000**	1,393,000	
thereof profit allocable to minority shareholders	(12)	– 10,519	– 3,445	
thereof profit allocable to hybrid capital investors	(13)	55,556	28,451	
thereof profit allocable to shareholders of Porsche AG	(13)	4,196,963	1,367,994	
Earnings per ordinary share from continiung operations (diluted and basic)	(13)	**239.80**	73.66	
Earnings per ordinary share from discontinued operations (diluted and basic)	(13)	**0.00**	4.44	
Earnings per preference share from continiung operations (diluted and basic)	(13)	**239.86**	73.72	
Earnings per preference share from discontinued operations (diluted and basic)	(13)	**0.00**	4.50	

* adjusted

Source: Porsche, Annual Report 2006/2007.

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Appendix C

Breakdown of Operating Income and Operating Expenses (€Thousands)

(3) Other operating income

Other operating income breaks down as follows:

	2006/07 T€	2005/06 T€
Income from stock options	6,926,751	767,169
Income from the reversal of imparments and provisions	72,988	47,538
Exchange rate gains	7,090	9,373
Sundry operating income	257,587	221,047
	7,264,416	**1,045,127**

Income from stock options mainly results from hedges transactions with cash compensation. Sundry operating income includes income from securities of T€ 8,855 (previous year: T€ 0) which was accounted for at fair value through profit or loss.

(6) Other operating expenses

Other operating expenses consist of:

	2006/07 T€	2005/06 T€
Stock options	3,333,474	506,139
Advertising	252,994	249,089
Selling and general administrative expenses	157,642	159,079
Dues, charges, fees, legal and advisory costs	88,903	34,789
Exchange rate losses	73,736	66,413
Repairs and maintenance	65,255	68,425
Rent and leases	31,136	30,408
Sundry operating expenses	596,959	594,976
	4,600,099	**1,709,318**

Expenses from stock options mainly results from hedges transactions with cash compensation. Sundry operating expenses includes expenses from securities of T€ 37 (previous year: T€ 0) which were accounted for at fair value through profit or loss.

Source: Porsche, Annual Report 2006/2007.

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Appendix D

Breakdown of Noncurrent and Current Other Receivables and Assets (€ Thousands)

(22) Non-current and current other receivables and assets

	July 31, 2007 T€	July 31, 2006 * T€
Derivative financial instruments	5,556,490	1,294,132
Other receivables	310,475	259,064
Prepaid expenses	23,139	19,451
	5,890,104	**1,572,647**
thereof non-current	285,662	172,659
thereof current	5,604,442	1,399,988

* the current other receivables decreased by T€ 1,306 in comparison to the previous year due to the transfer of the receivables of taxes on income.

The item derivative financial instruments mainly includes forward exchange contracts, currency options, stock options with cash compensation and compounders. A fractional amount of T€ 268,600 is due in more than one year (previous year: T€ 157,542).
Other assets mainly contain other taxes and advance payments; of these an amount of T€ 14,892 (previous year: T€ 12,853) is due in more than one year.

Prepaid expenses of T€ 23,139 (previous year: T€ 19,451) are principally attributable to the cut-off of other rent and marketing expenses as well as maintenance for hardware and software. Prepaid expenses include accruals and deferrals of T€ 2,170 due in more than one year (previous year: T€ 2,264).

The fair values of other receivables and financial assets correspond essentially to the carrying amounts. There were no significant valuation allowances for the other receivables and assets disclosed.

The financial assets held for trading have a carrying amount of T€ 5,091,886 (previous year: T€ 910,111).

Source: Porsche, Annual Report 2006/2007.

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Appendix E

Maturity of Derivative Financial Instruments Held by Porsche, July 31, 2007 (€Thousands)

		Nominal volume T€	Due within one year T€	Due within one to five years T€	Due within more than five years T€
	July 31, 2007				
Assets	Currency hedge	12,198,361	2,584,401	8,899,674	714,286
	thereof purchase of foreign currencies	174,595	89,980	84,615	–
	thereof sale of foreign currencies	12,023,766	2,494,421	8,815,059	714,286
	Interest hedge	1,091,319	54,313	687,006	350,000
	Stock options	10,553,364	10,553,364	–	–
		23,843,044	**13,192,078**	**9,586,680**	**1,064,286**
Equity and liabilities	Currency hedge	1,357,879	1,287,065	70,814	–
	thereof purchase of foreign currencies	718,647	718,647	–	–
	thereof sale of foreign currencies	639,232	568,418	70,814	–
	Interest hedge	902,830	–	592,770	310,060
	Stock options	13,473,485	13,473,485	–	–
		15,734,194	**14,760,550**	**663,584**	**310,060**

An amount of T€ 87,590 was reclassified in the fiscal year from the reserve for cash flow hedges to the income statement (previous year: T€ 155,266). The gain on the disposal of available-for-sale securities totaled T€ 15,233 (previous year: T€ 10,955), while the loss on the disposal came to T€ 0 (previous year: T€ 2,477).

Source: Porsche, Annual Report 2006/2007.

Appendix F

Zero-Coupon Yield Curves on November 30, 2007

	Maturity (years)									
	1	2	3	4	5	6	7	8	9	10
Euro (German Govt. Bonds)	3.94	3.81	3.80	3.84	3.90	3.96	4.03	4.09	4.15	4.21
US$ (US Treasuries)	3.19	3.04	3.08	3.23	3.43	3.62	3.80	3.97	4.10	4.21

Source: European Central Bank and Federal Reserve Board.

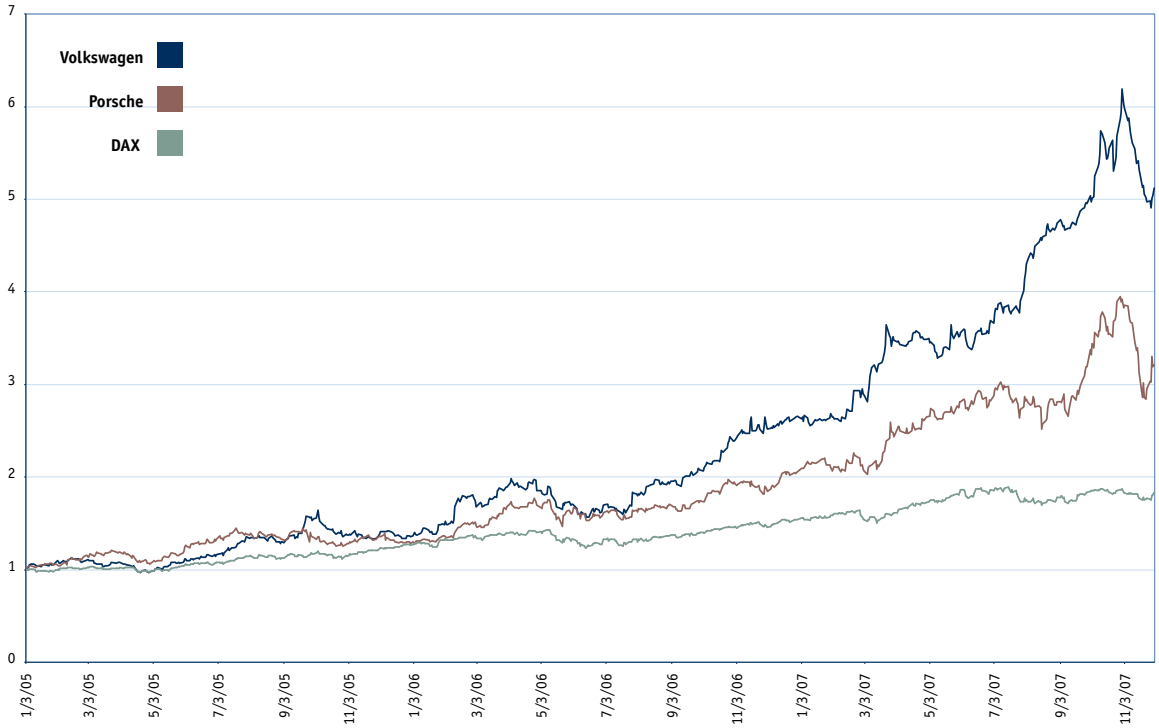
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Appendix G

Share Prices of Volkswagen and Porsche

	July 31, 2007	November 30, 2007
Volkswagen (ordinary shares)	132.54	163.84
Porsche (preferred shares)	1341.00	1504.10



Note: Volkswagen ordinary shares, Porsche preferred shares, and DAX Index (German stock market index), all rebased to 1.00 at the start of 2005, and adjusted for dividends and splits.

Source: DataStream.

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